

Safe Browsing Practices -- Guided Notes ANSWER KEY

Unit tp-6.12 | CompTIA Network+ N10-009 Obj. FC0-U71 6.2 | N10-008 FC0-U71 6.2

For instructor use / distribute after students complete the notes.

Question 1

HTTPS (HyperText Transfer Protocol Secure) encrypts communication between the browser and web server using _____. A _____ padlock in the browser address bar indicates an HTTPS connection. HTTPS does NOT mean the website is _____ -- it only means the _____ is encrypted. Phishing sites routinely use HTTPS.

Answer: TLS (SSL/TLS); padlock; trustworthy (safe); connection.

Question 2

A browser warning 'Your connection is not private' means the site's _____ certificate is invalid, expired, or not signed by a trusted _____ Authority. Users should not click 'proceed anyway' on unfamiliar sites because HTTPS protects the _____, not the site's _____.

Answer: SSL/TLS; certificate; transport; legitimacy (content).

Question 3

Drive-by downloads occur when visiting a _____ website causes malware to be automatically _____ without user interaction. They often exploit unpatched vulnerabilities in the _____ or _____. Keeping both updated and enabling _____ -scripting defenses (blocking unknown JavaScript) reduces risk.

Answer: Malicious (compromised); downloaded (installed); browser; plugins (extensions); no.

Question 4

URL inspection before clicking: check that the _____ domain is the expected one (attackers use similar-looking domains like paypa1.com). Look for _____ (misspellings designed to look like legitimate domains). Hover over links before clicking to see the _____ URL in the status bar. Shortened URLs (bit.ly) can _____ the real destination.

Answer: Actual; typosquatting; real; hide.

Question 5

Browser extensions can be a significant _____ risk. A malicious extension can read and modify _____ on pages visited, steal _____, and bypass _____ controls. Best practice: install only extensions from _____ sources with a high number of verified reviews, and remove unused extensions.

Answer: Security; content; credentials; browser; trusted (official store).

Question 6

Cookies are small files websites store in the browser to remember _____ and preferences. Third-party _____ track users across websites for advertising. _____ mode (Incognito) prevents

local storage of browsing history and reduces some tracking but does NOT make browsing _____ from the network or websites.

Answer: Sessions (login state); cookies; private; invisible.

Question 7

An ad blocker can reduce _____ ad attacks (malicious ads that deliver malware). A DNS _____ service (like Cloudflare 1.1.1.1 with filtering or OpenDNS) blocks requests to known _____ domains before they load. A Web proxy or _____ Filter categorizes and blocks unsafe websites at the network level.

Answer: Malvertising; filtering; malicious; Content.

Question 8 -- Real World Application

REAL WORLD: A student clicks a link in a Discord message that says 'free Steam games.' The link opens a page that looks identical to the Steam login page, but the URL is 'stearn.io' instead of 'store.steampowered.com'. They enter their username and password. Two hours later their Steam account is emptied of \$300 in items. Explain what attack occurred and list four specific warning signs the student should have caught.

Answer: Attack: credential harvesting phishing via a spoofed (fake) login page. The attacker created a pixel-perfect copy of the Steam login page on a lookalike domain. When the student entered credentials, they were sent directly to the attacker. The attacker then used the credentials on the real Steam site before the student noticed. Warning sign 1 -- Suspicious source: the link came through Discord DM from someone they may not know well, offering something too good to be true ('free games'). Unsolicited offers are a primary phishing vector. Warning sign 2 -- Wrong domain: 'stearn.io' is not 'steampowered.com'. The student should have looked at the address bar before entering any credentials. The legitimate URL would be store.steampowered.com (or steampowered.com). Warning sign 3 -- No HTTPS padlock from a trusted authority/wrong certificate: the padlock may exist but the certificate would be for 'stearn.io', not Valve/Steam. Warning sign 4 -- Urgency to log in: phishing pages want you to log in quickly before you notice something is wrong. Slowing down and inspecting before entering credentials is the most effective countermeasure.